Since $(A \cap B) \subset B$,,

$$B = (A \cap B) \cup (B - (A \cap B))$$

The events on the right hand side are disjoints. Hence by axiom 3

$$P(B) = P(A \cap B) + P(B - (A \cap B))$$

i.e. $$P[B - (A \cap B)] = P(B) - P(A \cap B) \qquad (8.2)$$

Substituting (8.2) in (8.1), it follows that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Corollary: If A, B and C are any three events, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Example 8.12

In the Annual sports meet, among the 260 students in XI standard in the school, 90 participated in Kabadi, 120 participated in Hockey, and 50 participated in Kabadi and Hockey. A Student is selected at random. Find the probability that the student participated in (i) Either Kabadi or Hockey, (ii) Neither of the two tournaments, (iii) Hockey only, (iv) Kabadi only, (v) Exactly one of the tournaments.

Solution:

$n(s)=260$

Let A : the event that the student participated in Kabadi

B : the event that the student participated in Hockey.

$n(A) = 90$; $n(B) = 120$; $n(A \cap B) = 50$

$$P(A) = \frac{n(A)}{n(S)} = \frac{90}{260}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{120}{260}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{50}{260}$$

(i) The probability that the student participated in either Kabadi or Hockey is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{90}{260} + \frac{120}{260} - \frac{50}{260} = \frac{160}{260} = \frac{8}{13}$$